



School of Engineering

Course: Diploma in Electronic and Computer Engineering

Module: EGE351 Automation Systems & Control

Name: _____

Module Group: _____

Team Members: _____

Date Submitted: _____

EXPERIMENT NO.: EGE351 / LAB 6

EXPERIMENT TITLE: Sequential Function Chart & Ladder Programming

OBJECTIVES:

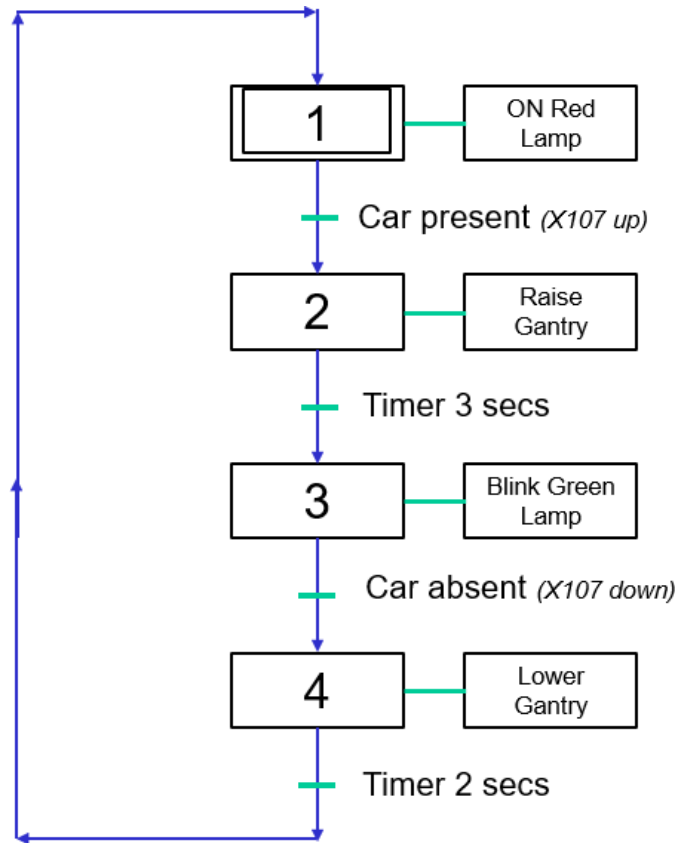
- a) Understand how to convert from PLC flowchart to ladder diagram
- b) Understand how to program and run the sequential program
- c) Know the two parts of PLC flowchart – sequence and output

EQUIPMENT: Programmable Logic Controller PLC FP7

COMPONENTS: Input/Output devices (Panel Box)

a) Sequential Programming

- 1) Convert the following PLC flowchart into PLC program or ladder diagram.



- 2) Download the PLC program (or ladder diagram) and test the system above.
- 3) Modify the PLC flowchart to also blink the Red Lamp at Step 4. Download & test.
- 4) Modify the PLC flowchart to also blink the Green Lamp when the Gantry is raised. Download & test.
- 5) Use a counter to count down as the Green Lamp is blinking at Step 3. The counter will be reset by PB1 and preset value is 30. The number will show on the monitor.